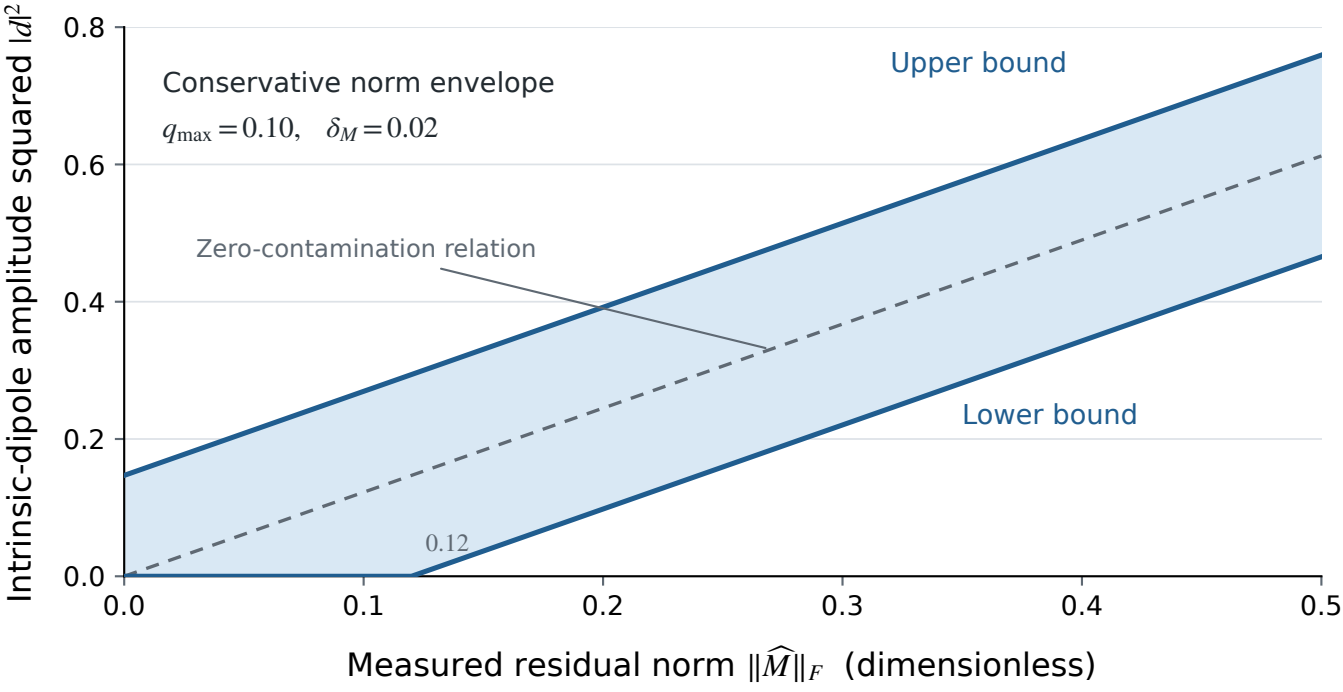
Conditional intrinsic-dipole bounds

Equation B7a | Analytical illustration; no observed sky data



$$\sqrt{3/2}\,\max(0, \|\widehat{M}\|_F - q_{\max} - \delta_M) \le |d|^2 \le \sqrt{3/2}\,(\|\widehat{M}\|_F + q_{\max} + \delta_M)$$

$M = K - \mathrm{STF}(aa^T)$; $\|\cdot\|_F$ is the full Cartesian Frobenius norm.

Assumptions: $\|Q_{\mathrm{intrinsic}}\|_F \le q_{\max}$ and $\|\widehat{M} - [Q_{\mathrm{intrinsic}} - \mathrm{STF}(dd^T)]\|_F \le \delta_M$.
Shading gives necessary scalar bounds, not full tensor compatibility or a confidence region.
All values are dimensionless. The illustrative range demonstrates the finite algebra only.